

CSR Creation Guide (Multi-SAN with SHA256)

This guide covers the complete process for creating a Certificate Signing Request (CSR) using MMC with the required Subject details, 16 SAN entries, and SHA256 configuration. Use the placeholders to insert screenshots where needed.

Step 1: Initialize the Request in MMC

- Press **Win + R**, type **mmc**, and press Enter.
- Go to **File** → **Add/Remove Snap-in**.
- Select **Certificates** → **Add** → **Computer account** → **Local computer** → **Finish** → **OK**.
- Expand **Certificates (Local Computer)** → **Personal**.
- Right-click **Certificates** → **All Tasks** → **Advanced Operations** → **Create Custom Request**.
- Click **Next**, select **Proceed without enrollment policy**, then click **Next**.
- Select **(No template) PKCS #10** and click **Next**.

■ *Insert Screenshot Here*

Step 2A: Fill Identity (Subject Tab)

- **Common Name (CN):** mypam.nisource.net
- **Organization (O):** NiSource (or your company name)
- **Organizational Unit (OU):** Security or IT
- **Locality (L):** Your City
- **State (S):** Indiana (full name)
- **Country (C):** US (2-letter code)

■ *Insert Screenshot Here*

Step 2B: Add Subject Alternative Names (SAN)

- DNS: mypam.nisource.net (must be included)
- DNS: PVWIDC1IFSCG003
- DNS: PVWIDC1IFSCG003.na.nisource.net
- DNS: PVWIDC2IFSCG003
- DNS: PVWIDC2IFSCG003.na.nisource.net
- DNS: PCWVDC1WAWBC007
- DNS: PCWVDC1WAWBC007.na.nisource.net
- DNS: PCWVDC1WAAPC106
- DNS: PCWVDC1WAAPC106.na.nisource.net
- DNS: PCWVDC2WAAPC011
- DNS: PCWVDC2WAAPC011.na.nisource.net
- IP: 10.96.129.127
- IP: 10.103.6.39
- IP: 10.235.21.17
- IP: 1.235.21.8
- IP: 10.235.133.54

■ *Insert Screenshot Here*

Step 3: Set Friendly Name (General Tab)

- **Friendly name:** mypam.nisource.net
- **Description:** CyberArk Multi-SAN Certificate for PVWA/PSM

■ *Insert Screenshot Here*

Step 4: Configure Hash and Keys (Private Key Tab)

- Cryptographic Provider: **Microsoft Software Key Storage Provider**
- Key size: **2048**
- Enable: **Make private key exportable**
- Hash Algorithm: **SHA256**

■ *Insert Screenshot Here*

Step 5: Save and Export

- Click **OK** to close Properties, then click **Next**.
- File Name: **C:\Certs\mypam_request.csr**
- File Format: **Base 64**
- Click **Finish**.

■ *Insert Screenshot Here*

PowerShell CSR Verification Script

Use the following PowerShell script to validate the CSR contents including Subject, SAN entries, key length, and hash algorithm.

```
param(
[Parameter(Mandatory=$true)]
[string]$CsrPath
)

Write-Host "=== CSR VERIFICATION START ===" -ForegroundColor Cyan

if (!(Test-Path $CsrPath)) {
Write-Error "CSR file not found: $CsrPath"
exit 1
}

$csr = New-Object
System.Security.Cryptography.X509Certificates.X509Certificate2
$csr.Import($CsrPath)

Write-Host "`nSubject:" -ForegroundColor Yellow
$csr.Subject

Write-Host "`nSignature Algorithm:" -ForegroundColor Yellow
$csr.SignatureAlgorithm.FriendlyName

Write-Host "`nKey Length:" -ForegroundColor Yellow
$csr.PublicKey.Key.KeySize

Write-Host "`nExtensions (look for SAN):" -ForegroundColor Yellow
foreach ($ext in $csr.Extensions) {
Write-Host $ext.Oid.FriendlyName
}

Write-Host "`n=== CSR VERIFICATION COMPLETE ===" -ForegroundColor Cyan
```